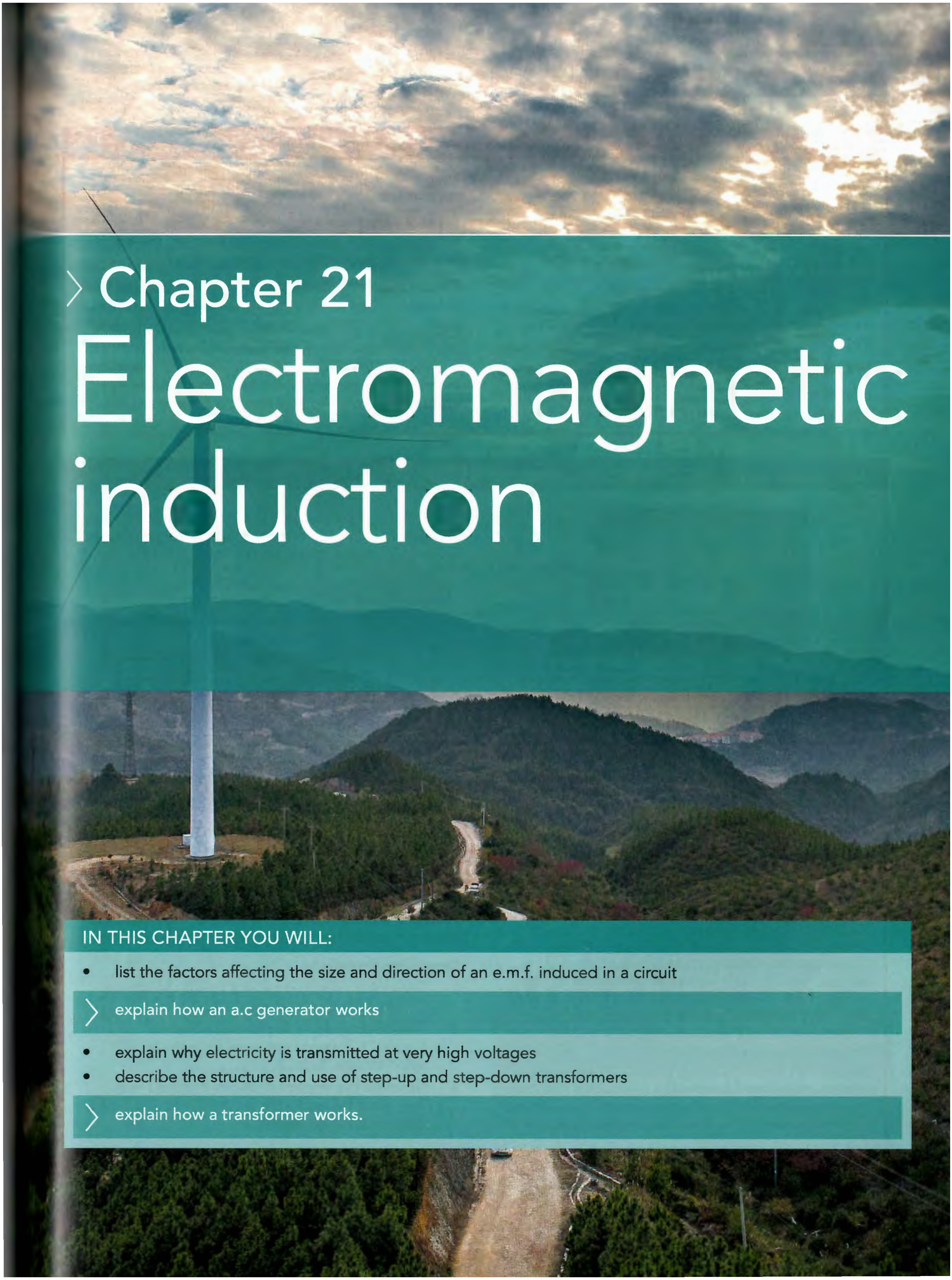


Chapter 21

Electromagnetic induction

IN THIS CHAPTER YOU WILL:

- list the factors affecting the size and direction of an e.m.f. induced in a circuit
- explain how an a.c generator works
- explain why electricity is transmitted at very high voltages
- describe the structure and use of step-up and step-down transformers
- explain how a transformer works.

GETTING STARTED

In Chapter 18 you learnt about the electrical quantities we can measure or calculate. These include: current, potential difference, resistance, charge, energy, power and electro-motive force.

For each quantity, write a definition, the letter used to represent the quantity in an equation (for example, *l* for length), and its unit.

Write down all the equations you know which involve these quantities.

In a small group, play a game to check your learning:

Create a set of cards. There should be one card for the name of each quantity, one card for the unit of each quantity, one card for time, and one card for seconds.

Place the cards face down in a pack. The first player turns two cards over. If they can connect the cards in a sentence or equation, they score a point.

Return the cards to the pack, mix them up, and then the next player takes their turn.

For example, if a player picks 'ohm' and 'current', they could say: 'when the resistance in ohms increases, the current drops'. If a student makes an incorrect link, other players can score a point by spotting the mistake and giving a correct answer.

WHY MUST ELECTRIC GUITARS HAVE METAL STRINGS?

Figure 21.1: The metal circles (on the white bar below the strings of this electric guitar) pick up sound vibrations. Notice that the strings do not touch the pickup.

We saw in Chapter 20 that a current flowing in a magnetic field experiences a force which can make it move. In this chapter we will investigate the reverse process. When a conductor and magnetic field move relative to each other, a current flows in the conductor. This is the principle behind the operation of power stations.

This effect is also used to pick up sound vibrations in an electric guitar. The guitar strings are magnetised by a permanent magnet in the pick-up below the strings. This is why they must be made of a magnetic metal such as nickel or steel. The pick-up device – the white bar below the strings in Figure 21.1 – also contains coils of copper wire which are wrapped around the permanent magnets. The strings vibrate when the guitar is played. The amplitude and frequency of these vibrations depend on the note being played. We now have relative movement between a conductor (the coils) and a field (around the magnetised strings) and so a current flows in the coils. This current is amplified and fed to a loudspeaker. The size and frequency of the current depends on the vibrations of the strings. This allows music to be played.

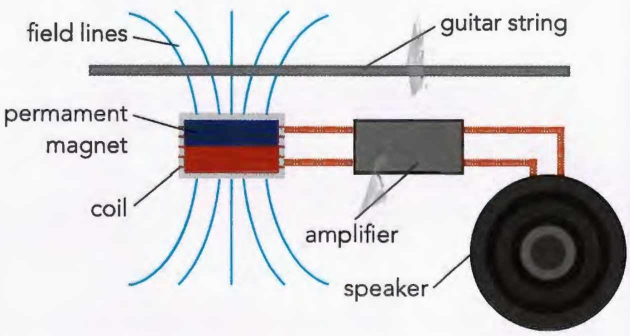


Figure 21.2: How the pickup in a guitar works.

Figure 21.2 shows the field lines from the permanent magnet which magnetises the metal strings. The changing current generated in the coil is passed to the speaker which contains a magnet. As the changing current flows, the speaker cone moves, producing sound – this is the motor effect.

Discussion questions

1. The guitar pickup does not touch the strings. Explain why this is an advantage.
2. The electromagnetic pickup shown in Figure 21.2 would not work for a guitar with nylon strings. Explain why.
3. Describe the energy changes involved from the guitar being strummed to the sound being heard from the speaker.

21.1 Generating electricity

A motor is a device that transfers energy by an electrical current into a mechanical (kinetic) energy store. An electrical generator does the opposite – it transfers energy from a mechanical energy store by an electrical current. An electric motor can be used in reverse to generate electricity. If you connect an electric motor to a lamp and spin its axle, the lamp will light, showing that you have generated a voltage which causes a current to flow through the lamp (Figure 21.3).

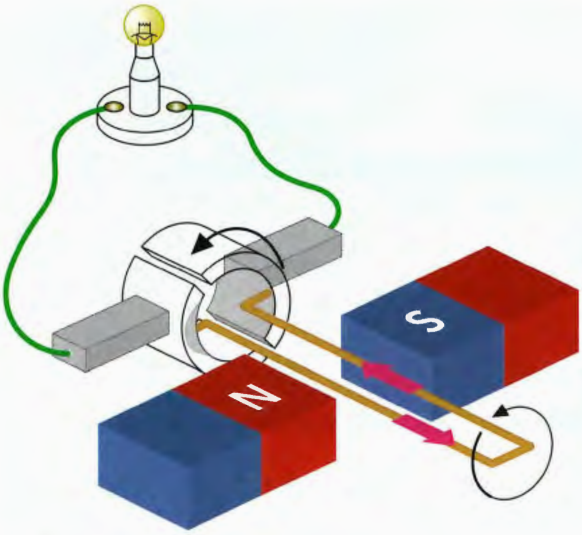


Figure 21.3: A motor can act as a generator. Spin the motor and the lamp lights, showing that an induced current flows around the circuit.

Inside the motor, the coil is spinning around in the magnetic field provided by the permanent magnets. The result is that a current flows in the coil, and this is shown by the lamp. We say that the current has been induced, and the motor is acting as a generator.

There are many different designs of generator, just as there are many different designs of electric motor. Generators in power stations supply electricity to communities. If you have a bicycle, you may have a generator of a different sort – a dynamo – for powering the lights.

Figure 21.4: The dynamo rubs against the wheel causing it to turn. This generates current to light the bicycle lamp.

The power station generators shown in Figure 21.10 generate alternating current at a voltage of about 25 kV. The turbines are made to spin by the high-pressure steam from the boiler. The generator is on the same axle as the turbine, so it spins too. A coil inside the generator spins around inside some fixed electromagnets, which provide the magnetic field. A large current is then induced in the rotating coil, and this is the current that the power station supplies to consumers.

All of these generators have three things in common:

- a magnetic field (provided by magnets or electromagnets)
- a coil of wire (fixed or moving)
- movement (the coil and magnetic field move relative to one another).

When the coil and the magnetic field move relative to each other, a current flows in the coil if it is part of a complete circuit. This is known as an induced current. If the generator is not connected up to a circuit, there will be an **induced e.m.f.** (or **induced voltage**) across its ends, ready to make a current flow around a circuit.

> **KEY WORDS**
>
> **induced e.m.f.:** (or induced voltage) the e.m.f. created in a conductor when it cuts through magnetic field lines

The principles of electromagnetic induction

The process of generating electricity from motion is called **electromagnetic induction**. The science of electromagnetism was largely developed by Michael Faraday. He invented the idea of the magnetic field and drew field lines to represent it. He also invented the first electric motor. Then he extended his studies to show how the motor effect could work in reverse to generate electricity. In this section, we will look at the principles of electromagnetic induction that Faraday discovered.

As we have seen, a coil of wire and a magnet moving relative to each other are needed to induce a voltage across the ends of a wire. If the coil is part of a complete circuit, the induced e.m.f. will make an induced current flow around the circuit.

In fact, you do not need to use a coil – a single wire is enough to induce an e.m.f., as shown in Figure 21.5a. The wire is connected to a sensitive meter to show when a current is flowing.

- Move the wire down between the poles of the magnet and a current flows.
- Move the wire back upwards and a current flows in the opposite direction.
- Alternatively, the wire can be kept stationary and the magnet moved up and down. Again, a current will flow.

You can see similar effects using a magnet and a coil (Figure 21.5b). Pushing the magnet into and out of the coil induces a current, which flows back and forth in the coil. Here are two further observations:

- Reverse the magnet to use the opposite pole and the current flows in the opposite direction.
- Hold the magnet stationary next to the wire or coil and no current flows. They must move relative to each other, or nothing will happen.

In these experiments it helps to use a centre-zero meter. Then, when the needle moves to the left, it shows that the current is flowing one way; when it moves to the right, the current is flowing the other way.

KEY WORDS

electromagnetic induction: the production of an e.m.f across an electrical conductor when there is relative movement between the conductor and a magnetic field

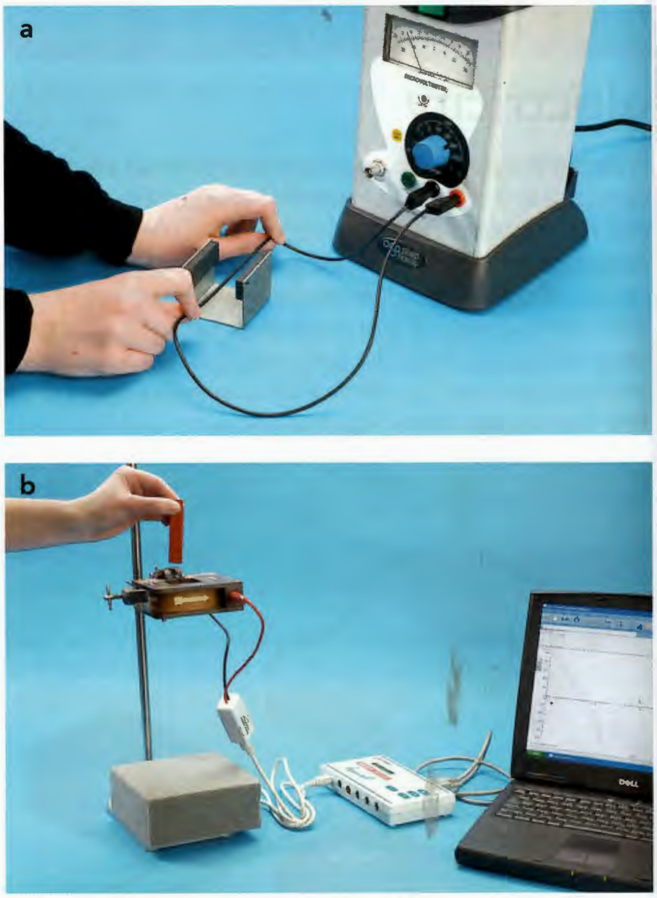


Figure 21.5a: Move a wire up and down between the poles of a stationary magnet and an induced current will flow.
b: Similarly, move a magnet into and out of a coil of wire and an induced current will again flow.

Increasing the induced e.m.f.

There are three ways to increase the e.m.f. induced in a coil or wire:

- use a stronger magnet
- move the wire or coil more quickly relative to the magnet
- use a coil with more turns of wire. Each turn of wire will have an e.m.f. induced in it, and these all add together to give a bigger e.m.f.

EXPERIMENTAL SKILLS 21.1

Inducing electricity

In this experiment you will induce an e.m.f. and investigate the factors which affect the induced e.m.f., including the factors affecting the size and direction of the e.m.f. induced when a conductor moves relative to a magnetic field.

Safety: Some sensitive meters have a locking mechanism which prevents them being damaged during movement. Check your meter before you begin.

Getting started

Look carefully at the meter you are using. Explain:

- why zero is at the centre of the scale
- how you can tell it is a very sensitive meter.

You will need:

- thin insulated wire
- strong bar magnet
- 2 magnadur magnets and a yoke
- sensitive, centre-zero ammeter or voltmeter.

Method

1 Coil 2.0 metres of thin insulated wire with bare ends to make a solenoid approximately 5 cm in diameter. The coil can be flat as shown in Figure 21.6 rather than long.

2 Connect the ends of the coil to the terminals of a sensitive voltmeter or ammeter.

3 Bring one pole of a bar magnet towards and into the centre of the coil. Observe the reading on the meter.

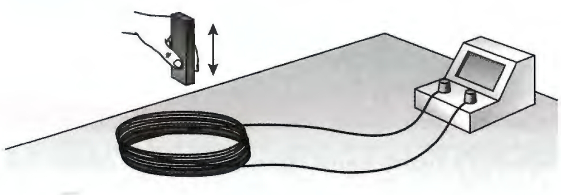


Figure 21.6: Experimental set-up.

4 Now investigate how the reading on the meter changes in different circumstances:
   - Move the bar magnet at different speeds into the coil.
   - Use the opposite pole of the bar magnet.
   - Move the bar magnet out of the coil.
   - Hold the bar magnet stationary at different distances from the coil.

5 Straighten out the wire. Keep the ends connected to the meter.

6 Mount two magnadur magnets on a yoke. Ensure that opposite poles are facing each other so that there is a strong magnetic field between the magnets.

7 Hold a section of the wire, approximately 10 cm in length, between your two hands. Move the wire downwards through the field (Figure 21.7). Observe the reading on the meter.

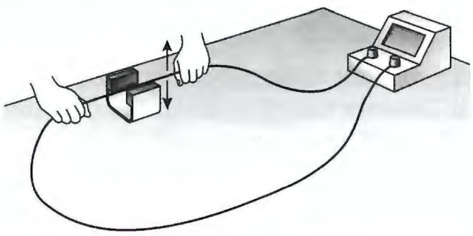


Figure 21.7: Experimental set-up (part 2)

8 Now investigate how the reading on the meter changes in different circumstances.
   - Move the wire at different speeds through the magnetic field.
   - Reverse the direction of the magnetic field.
   - Move the wire out of the magnetic field.
   - Hold the wire stationary in the magnetic field.

CONTINUED

Question

1 Copy and complete these sentences to record your observations:

An e.m.f. is induced in a conductor in a magnetic field if either the ______ or the ______ move across the other.

The size of the e.m.f. induced in a coil can be increased by increasing ______, ______ or ______.

The direction of the e.m.f. can be ______ by reversing the field or the direction of movement.

Questions

1 Which of the following would not induce a current:
   - **A** moving a wire between the poles of a magnet
   - **B** taking a bar magnet out of a coil or wire
   - **C** holding a wire between the poles of a magnet
   - **D** spinning a coil or wire in a magnetic field.

2 Marcus holds a piece of copper wire which is connected to an ammeter between the poles of a magnet, but he does not get a reading on the ammeter. What should he do to make a current flow?

3 A student moves a bar magnet into a coil of wire and a current flows. Describe two changes which would make the current flow in the opposite direction.

4 An a.c generator and a cell can both be used to light a bulb. Describe how the current flowing through the bulb is different in each case.

Induction and field lines

We can understand electromagnetic induction by thinking about magnetic field lines. Figure 21.8 shows the field lines between the poles of a horseshoe magnet. As the wire is moved down between the poles of the magnet, it cuts the field lines of the magnet. Cutting the field lines induces the current.

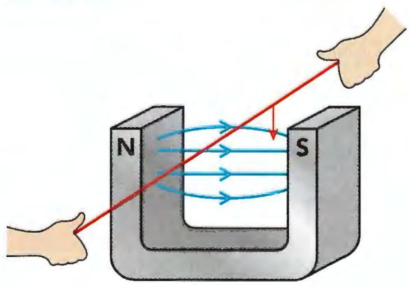


Figure 21.8: As the wire cuts through the field lines, an e.m.f. is induced in it. If the wire is part of a complete circuit, a current will flow.

This idea helps us to understand the factors that affect the magnitude and direction of the induced e.m.f.

- When the magnet is stationary, there is no cutting of field lines and so no e.m.f. is induced.
- When the magnet is further from the wire, the field lines are further apart and so fewer are cut, giving a smaller e.m.f.
- When the magnet is moved quickly, the lines are cut more quickly and a bigger e.m.f. is induced.
- A coil gives a bigger effect than a single wire, because each turn of wire cuts the magnetic field lines. This means that each contributes to the induced e.m.f.

Fleming's right-hand rule

We have seen that, when a wire is moved so that it cuts across a magnetic field, a current is induced in the wire. How can we work out the direction of the current? In Chapter 20, we saw the opposite effect. When a current flows in a magnetic field, there is a force on it so that it moves. The directions of force, field and current were given by Fleming's left-hand rule. It is not surprising to find that, in the case of electromagnetic induction, the directions are given by **Fleming's right-hand rule**. As shown in Figure 21.9, the thumb and first two fingers show the directions of the same quantities. Take care to use the correct hand.

KEY WORDS

Fleming's right-hand rule: a rule that gives the relationship between the directions of force, field and current when a current flows across a magnetic field

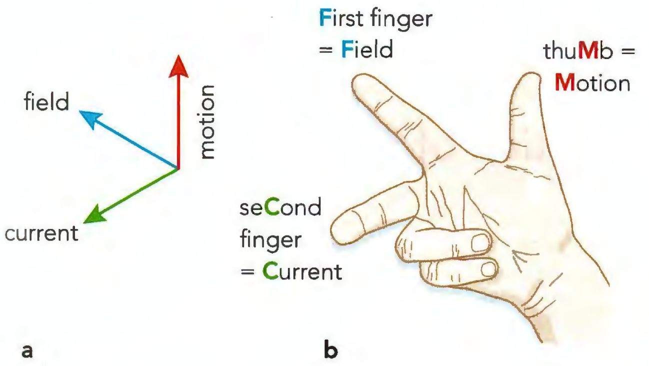


Figure 21.9a: When a current is induced in a wire, motion, field and current are at right angles to each other. **b:** Fleming's right-hand rule is used to work out the direction of the induced current.

An a.c. generator

Faraday's discovery of electromagnetic induction led to the development of the electricity supply industry. In particular, it allowed engineers to design generators that could supply electricity. At first, this was only done on a small scale, but gradually generators got bigger and bigger, until, like the ones shown in Figure 21.10, they were capable of supplying the electricity demands of thousands of homes.

Figure 21.10: The turbine and generator in the generating hall of a nuclear power station. The turbines are fed by high-pressure steam in pipes.

A generator of this type produces alternating current (a.c.). Alternating current flows first one way then the other as the coils turn in the magnetic field.

Figure 21.11 shows a simple **a.c. generator**, which produces alternating current. In principle, an a.c generator is like a d.c. motor, working in reverse. The axle is made to turn so that the coil spins around in the magnetic field, and a current is induced. The other difference is in the way the coil is connected to the circuit beyond. A d.c. motor uses a split-ring commutator, whereas an a.c. generator uses **slip rings**. The slip rings rotate with the coil. The brushes rub against the slip rings and so have the same e.m.f. as the sides of the coil.

KEY WORDS

a.c. generator: a device such as a dynamo used to generate alternating current

slip rings: a device used to allow current to flow to and from the coil of an a.c. generator

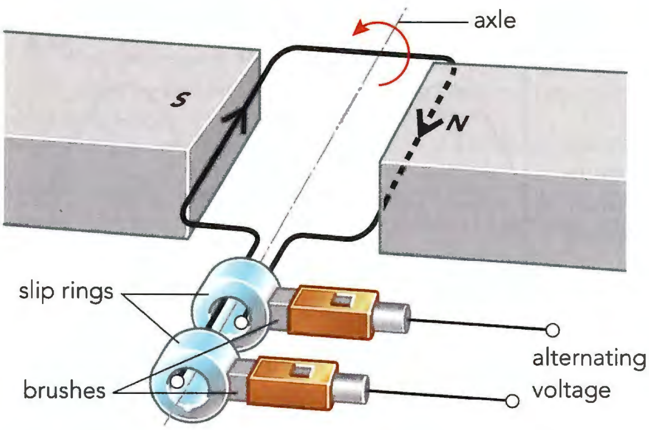


Figure 21.11: A simple a.c. generator works like a motor in reverse. The slip rings and brushes are used to connect the alternating current to the external circuit.

Why does this generator produce alternating current? As the coil rotates, each side of the coil passes first the magnetic north pole and then the south pole.

Figure 21.12 shows a graph of this. When the coil is horizontal, it cuts through the field lines inducing a voltage. As it turns to vertical, it cuts fewer field lines so the voltage decreases to zero. As it continues back to horizontal it cuts through the field lines in the opposite direction, giving a peak voltage in the opposite direction.

This means that the induced current flows first one way, and then the other. In other words, the current in the coil is alternating.

The current flows out through the slip rings. Each ring is connected to one end of the coil, so the alternating current flows out through the brushes, which press against the rings.

There are four ways of increasing the voltage generated by an a.c. generator like the one shown in Figure 21.11:

- turn the coil more rapidly
- use a coil with more turns of wire
- use a coil with a bigger area
- use stronger magnets.

Each of these changes increases the rate at which magnetic field lines are cut, and so the induced e.m.f. is greater. For the a.c. generator shown in Figure 21.11, each revolution of the coil generates one cycle of alternating current. Spin the coil 50 times each second and the a.c. generated has a frequency of 50 Hz.

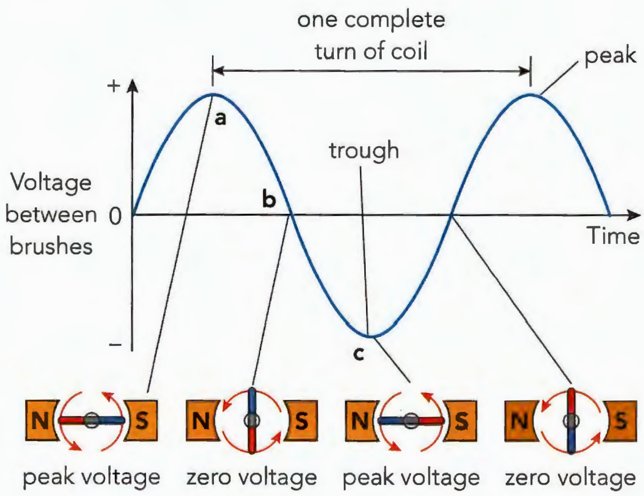


Figure 21.12: A graph to represent an alternating voltage. As the coil turns, the voltage reaches a peak in one direction, decreases to zero, then reverses.

Figure 21.12 shows how the e.m.f. produced by an a.c. generator varies as the coil turns. If we think about how the coil cuts through magnetic field lines, we can understand why the a.c. graph varies between positive and negative values. With the coil in the horizontal position, as shown in position a, its two long sides are cutting rapidly through the magnetic field lines. This gives a large induced e.m.f., corresponding to a peak in the a.c. graph. When the coil is vertical (position b), its long sides are moving along the field lines, so they are not cutting them. This gives no induced e.m.f., a zero point on the a.c. graph. When the coil has turned through 180° to position c, it will be cutting field lines quickly again, but in the opposite direction, so the induced e.m.f. will again be large, but this time it will be negative – this corresponds to a trough on the graph.

Direction of the induced e.m.f.

How is the direction of an induced current determined? The answer is that the current (like all currents) has a magnetic field around it. This field always pushes back against the field that is inducing the current. So, for the coil shown in Figure 21.13, when the magnet's north pole is pushed towards the coil, the current flows to produce a north pole at the end of the coil nearest the magnet. These two north poles repel each other. This means that you have to push the magnet towards the coil and that you have to do work. The energy you use in pushing the magnet is transferred to the current. That is where the energy carried by the current comes from. It comes from the work done in making a conductor cut through magnetic field lines.

An induced current always flows in such a way that its magnetic field opposes the change that causes it. This is known as **Lenz's law**.

KEY WORDS

Lenz's law: the direction of an induced current always opposes the change in the circuit or the magnetic field that produces it

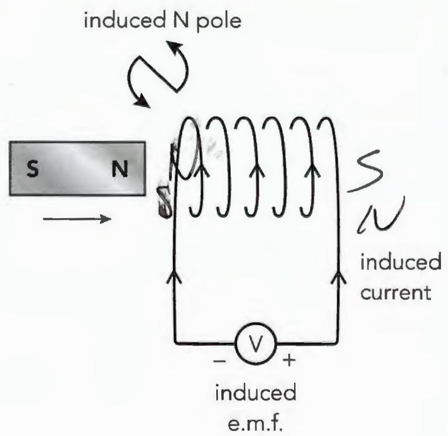


Figure 21.13: Consider a bar magnet being moved into a coil north pole first. Lenz's law tells us that the e.m.f. induced will oppose this. Therefore, the end the magnet approaches must become a north pole, to repel the approaching north pole. This means that the current must flow anticlockwise at that end of the coil.

Questions

5 A student moves the north pole of a magnet towards a coil of wire (as shown in Figure 21.13), so that an induced current flows. State two ways in which the student could cause a bigger induced current to flow.

6 Draw a diagram similar to Figure 21.13 to show what happens when the magnet is moved away from the coil. (Hint: remember, the induced current will oppose this change.)

ACTIVITY 21.1

Generating fitness

Figure 21.14: Using an exercise bike to charge a mobile phone.

The woman in Figure 21.14 is charging her mobile phone. As she pedals the exercise bike, she turns a generator to induce a current which is used to charge her phone.

Produce a poster to attract customers to use a bike charger in a public place. You should include information about how the system helps to:

- save money
- improve the user's health
- protect the environment.

You should also include a more detailed section for the user to read as they pedal. This should explain how the system works and advise the customer how to charge their phone as quickly as possible.

21.2 Power lines and transformers

Figure 21.15: Electricity is usually generated at a distance from where it is used. If you look on a map, you may be able to trace the power lines that bring electrical power to your neighbourhood.

Power stations may be 100 km or more from the places where the electricity they generate is used. This electricity must be distributed around the country.

High-voltage electricity leaves the power station. Its voltage may be as high as 1 million volts. To avoid danger to people, it is usually carried in cables called **power lines** slung high above the ground between tall pylons. Lines of pylons carry wires across the countryside, heading for the urban and industrial areas that need the power (Figure 21.16). This is a country's **national grid**.

KEY WORDS

power lines: cables used to carry electricity from power stations to consumers

national grid: the system of power lines, pylons and transformers used to carry electricity around a country

When the power lines approach the area where the power is to be used, they enter a local distribution centre. Here the voltage is reduced to a less hazardous level, and the power is sent through more cables (overhead or underground) to local substations. In the substation, the voltage is reduced to the local supply voltage, typically 230 V. Wherever you live, there is likely to be a substation

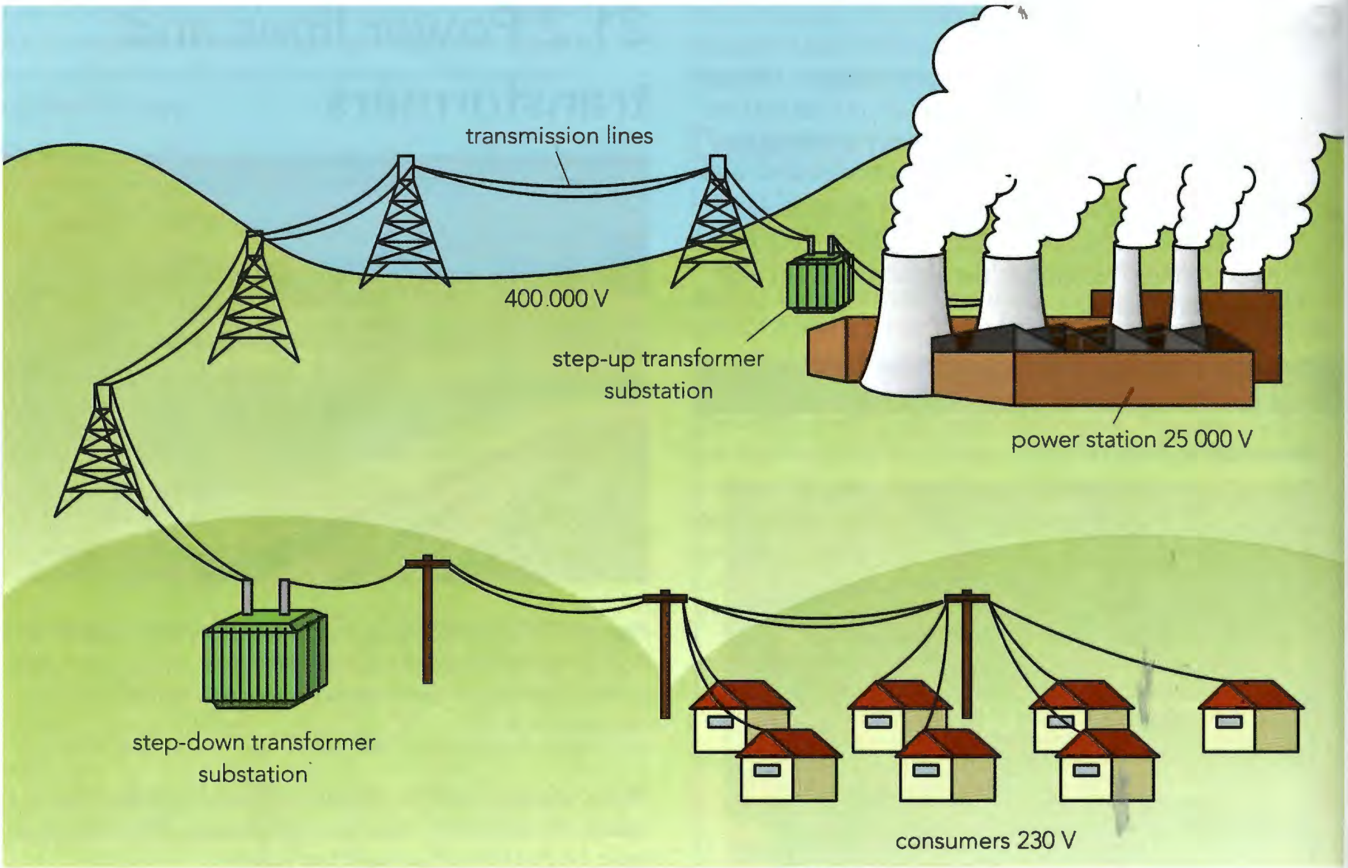


Figure 21.16: In a national grid, voltages are increased to reduce energy losses, then reduced to a relatively safe level.

in the neighbourhood. It may be in a securely locked building, or the electrical equipment may be surrounded by fencing, which carries notices warning of the hazard (Figure 21.17).

Figure 21.17: An electricity substation has warning signs like this to indicate the extreme hazard of entering the substation.

From the substation, electricity is distributed to houses, shops, etc. In some countries, the power is carried in cables buried underground. Other countries use tall poles and overhead wires.

Why use high voltages?

The high voltages used to transmit electrical power around a country are dangerous. That is why the cables that carry the power are supported high above people, traffic and buildings on tall pylons. Sometimes the cables are buried underground, but this is much more expensive, and the cables must be safely insulated. The reason for using high voltages is to reduce the loss of energy due to the cables heating up. This heating happens much more when the same power is transmitted at a lower voltage.

Transformers

A **transformer** is a device used to increase or decrease the voltage of an electricity supply. They are designed to be as efficient as possible (up to 99.9% efficient). This is because the electricity we use may have passed through as many as ten transformers before it reaches us from the power station. A loss of 1% of energy in each transformer would represent a total waste of 10% of the energy leaving the power station.

Power stations typically generate electricity at 25 kV. This has to be converted to the grid voltage (typically 400 kV) using transformers. This is known as stepping up the voltage.

Figure 21.18a shows the construction of a suitable transformer. Every transformer has three parts:

- A **primary coil**: the incoming voltage V_p is connected across this coil.
- A **secondary coil**: this provides the voltage V_s to the external circuit.
- An iron core: this links the two coils.

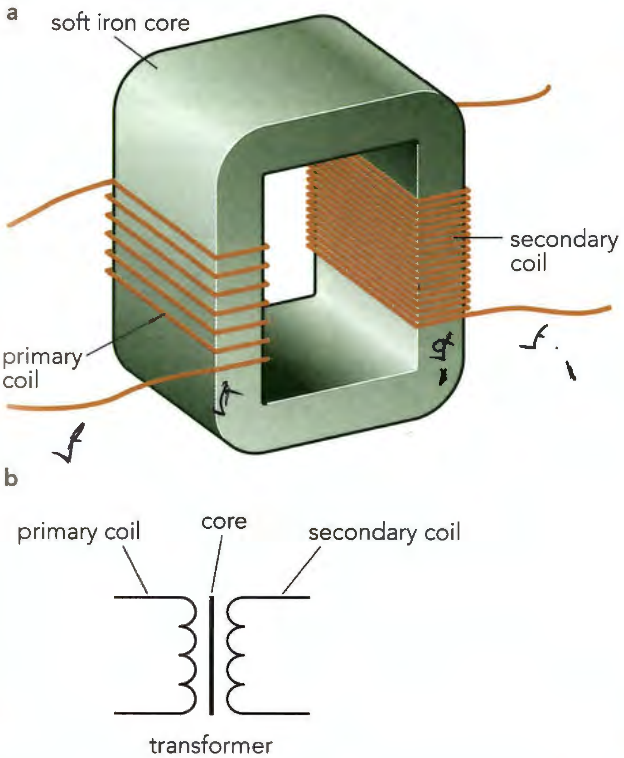


Figure 21.18a: The structure of a transformer. This is a step-up transformer because there are more turns on the secondary coil than on the primary. If the connections to it were reversed, it would be a step-down transformer.
b: The circuit symbol for a transformer shows the two coils with the core between them.

Notice that there is no electrical connection between the two coils. They are linked together only by the soft iron core. The wires are insulated so no current flows from one coil to the other. Notice also that the voltages are both alternating voltages. Transformers only work with a.c. All they do is change the size of an a.c. voltage.

The power station transformer described earlier steps up the voltage from 25 kV to 400 kV – it is increased by a factor of 16. To step up the voltage by a factor of 16, there must be 16 times as many turns on the secondary coil as on the primary coil. By comparing the numbers of turns on the two coils we can tell how the voltage will be changed.

- A **step-up transformer** increases the voltage. There are more turns on the secondary coil than on the primary coil.
- A **step-down transformer** reduces the voltage. There are fewer turns on the secondary coil than on the primary coil.

KEY WORDS

transformer: a device used to change voltage of an a.c. electricity supply

primary coil: the input coil of a transformer

secondary coil: the output coil of a transformer

step-up transformer: a transformer which increases the voltage of an a.c. supply

step-down transformer: a transformer which decreases the voltage of an a.c. supply

Note that, when the voltage is stepped up, the current is stepped down, and when the voltage is stepped down, the current is stepped up.

The ratio of the number of turns tells us the factor by which the voltage will be changed. We can write an equation, known as the transformer equation, relating the two voltages, V_p and V_s, to the numbers of turns on each coil, N_p and N_s:

KEY EQUATION

$$\frac{\text{voltage across primary coil}}{\text{voltage across secondary coil}} = \frac{\text{number of turns on primary}}{\text{number of turns on secondary}}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

WORKED EXAMPLE 21.1

There are very large transformers between power stations to the transmission cables.

One of these transformers has 800 turns on its primary coil and 16 000 turns on its secondary coil. The voltage across its primary coil is 25 kV.

a State what type of transformer this is.

b Calculate the voltage across its secondary coil.

Step 1: Write down what you know: The transformer has more turns on the secondary coil than the primary so it is a step-up transformer. This means the voltage should increase. Use this to check your answer to the next part.

Step 2: Draw a simple transformer as shown in Figure 21.19 and mark on it the information given in the question.

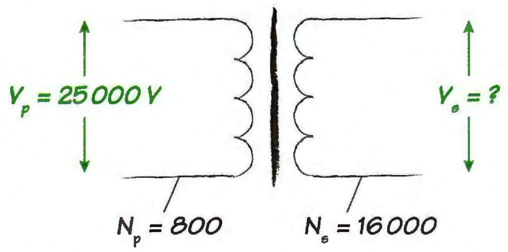


Figure 21.19: Transformer symbol with the quantities given in the question.

Step 3: Write down the transformer equation.

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

Step 4: Substitute in the values from the question.

$$\frac{25\,000\text{ V}}{V_s} = \frac{800}{16\,000}$$

Step 5: Rearrange the equation to find V_s.

$$V_s = \frac{25\,000\text{ V} \times 16\,000}{800}$$

$$= 500\,000\text{ V}$$

Check that this is greater than the primary voltage as expected.

Answer

a step-up transformer

b $V_s = 500\,000$ V

It is often useful to do a quick mental calculation first. For example, if a transformer decreases voltage from 230 V to 60 V, it has reduced it by about a quarter. This means the secondary coil must have roughly a quarter of the number of turns than those on the primary. This is an approximate calculation. It helps you check whether your final answer is sensible.

Questions

7 **a** Copy and complete these sentences.

Electricity is distributed by the national ______. This consists of substations, cables and ______.

The electricity is transmitted at very high ______ to ______ energy being lost as heat in the cables.

The voltage can be increased or decreased by ______.

b Figure 21.20 shows a transformer.

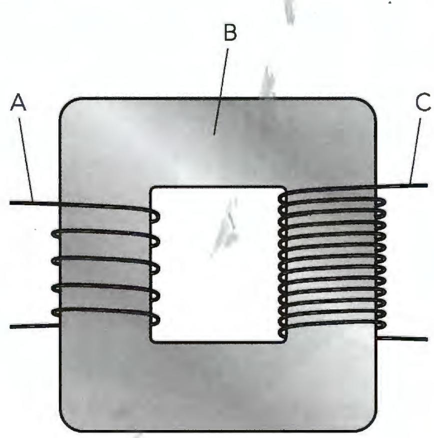


Figure 21.20: A transformer.

Name the parts labelled A, B and C.

c Explain whether this is a step-up or step-down transformer.

8 A transformer has ten turns on the primary coil and five turns on the secondary. A voltage of 12 V is applied across the primary coil.

a Is this a step-up or step-down transformer?

b Use the transformer equation to calculate the output voltage of the transformer.

9 Copy and complete Table 21.1. Calculate the missing information using the transformer equation.

N_p	N_s	V_p	V_s	Step-up or step-down transformer?
10	20		12	step-up
10		1.2	12	
	50	240	6	
10 000	20	115 000		

Table 21.1

10 A portable radio has a built-in transformer so that it can work from the mains instead of batteries. Is this a step-up or step-down transformer?

11 A transformer increases the 1100 V from a power station to 132 000 V for transmission. Calculate the number of turns on the primary coil when the secondary coil has 6000 turns.

REFLECTION

In these calculations it is useful to think about whether the answer is sensible. Which of the following suggestions are helpful for you? What does this tell you about how you work?

- Draw the transformer and write the numbers on it.
- Draw a diagram of the situation.
- Do a quick approximate calculation first.
- Decide whether the transformer is step-up or step-down, and what this tells you about your answer.

Think about whether you can apply any of these ideas when you do other calculations.

21.3 How transformers work

Transformers only work with alternating current (a.c.). To understand why this is, we need to look at how a transformer works (Figure 21.21). It makes use of electromagnetic induction.

The primary coil has alternating current flowing through it. It is, therefore, an electromagnet, and produces an alternating magnetic field. The core transports this alternating field around to the secondary coil. Now the secondary coil is a conductor in a changing magnetic field. A current is induced in the coil. This is another example of electromagnetic induction at work.

When the secondary coil has only a few turns, the e.m.f. induced across it is small. When it has a lot of turns, the e.m.f. will be large. Hence, to increase the voltage out, we need a secondary coil with many more turns than the primary coil.

When direct current is connected to a transformer, there is no output voltage. This is because the magnetic field produced by the primary coil does not change. With an unchanging field passing through the secondary coil, no voltage is induced in it.

Notice from Figure 21.21 that the magnetic field links the primary and secondary coils. The energy brought by the current in the primary coil is transferred to the secondary coil by the magnetic field. This means that the core must be very good at transferring magnetic energy. A soft magnetic material must be used – usually an alloy of iron with a small amount of silicon. (Recall that soft magnetic materials are ones that can be magnetised and demagnetised easily.)

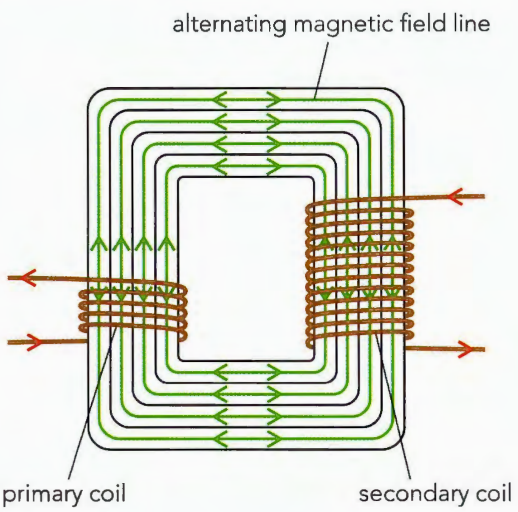


Figure 21.21: The a.c. in the primary coil produces a varying magnetic field in the core. This induces a varying current in the secondary coil.

Even in a well-designed transformer, some energy is lost because of the resistance of the wires, and because the core resists the flow of the changing magnetic field.

Questions

12 a What is the function of the core of a transformer?

b Why must the core be made of a soft magnetic material?

13 Explain why a transformer will not work with direct current.

Calculating current

To transmit a certain power, P, we can use a small current, I, if we transmit the power at high voltage, V. This follows from the equation for electrical power (see Chapter 18):

electrical power P = IV

Worked Example 21.2 shows how this works.

Energy saving

There is a good reason for using high voltages. It means that the current flowing in the cables is relatively low, and this wastes less energy. This can be explained as follows.

A consumer needs a certain amount of power. This power can be delivered as:

- high voltage, low current or
- low voltage, high current.

Calculating power losses

When a current flows in a wire or cable, some of the energy it is carrying is lost because of the cable's resistance – the cables get warm. A small current wastes less energy than a high current.

We can calculate the rate of energy loss in the cables, or the power loss, by putting together two equations we met in Chapter 18:

power = current × potential difference or $P = IV$

and

potential difference = current × resistance or $V = IR$

$P = IV$ becomes $P = I(IR)$ or $P = I^2R$.

KEY EQUATION

power loss = square of current in the cable × resistance

$$P = I^2R$$

Electrical engineers do everything they can to reduce the energy losses in the cables. If they can reduce the current to half its value (by doubling the voltage), the losses will be one-quarter of their previous value. This is because power losses in cables are proportional to the square of the current flowing in the cables:

- double the current gives four times the losses
- three times the current gives nine times the losses.

WORKED EXAMPLE 21.2

A 20 kW generator gives an output of 5 kV. This is transmitted to a workshop by cables with a resistance of 20 Ω. Calculate:

a the power loss in the cables

b the effect of using a transformer to increase the output voltage to 20 kV assuming that the power output remains the same.

When answering this question, it is important to realise there are two different powers involved – the power of the generator and the power loss in the cables.

Step 1: Calculate the current in the wires at 5 kV using the equation:

generator power, $P = IV$

Rearrange the equation and substitute values:

$I = P \div V = 20\,000\text{ W} \div 5000\text{ V} = 4\text{ A}$

Step 2: Calculate the power loss:

power loss, $P = I^2R = (4\text{ A})^2 \times 20\,\Omega = 320\text{ W}$ (this is the answer to part **a**).

Step 3: Calculate the current in the wires at 20 kV:

$P = IV$

$I = P \div V = 20\,000\text{ W} \div 20\,000\text{ V} = 1\text{ A}$

Step 4: Calculate the power loss:

$P = I^2R = (1\text{ A})^2 \times 20\,\Omega = 20\text{ W}$

Answer

a 320 W

b Using the transformer to decrease the voltage greatly reduces the power loss, from 320 W to 20 W.

From the results of Worked Example 21.2, you can see why electricity is transmitted at high voltages. The higher the voltage, the smaller the current in the cables, and so the smaller the energy losses. Increasing the voltage by a factor of four reduces the current by a factor of four. This means that the power lost in the cables is greatly reduced (in fact, it is reduced by a factor of 4^2, which is 16), and so thinner cables can safely be used.

The current flowing in the cables is a flow of coulombs of charge. At high voltage, we have fewer coulombs flowing, but each coulomb carries more energy with it.

Thinking about power

If a transformer is 100% efficient, no power is lost in its coils or core. This is a reasonable approximation, because well-designed transformers waste only about 0.1% of the power transferred through them. This allows us to write an equation linking the primary and secondary voltage, V_p and V_s, to the primary and secondary currents, I_p and I_s, using $P = IV$.

KEY EQUATION

power in to primary coil = power out of secondary coil

$$I_p \times V_p = I_s \times V_s$$

It is important to remember that this equation assumes that no power is lost in the transformer. Worked Example 21.3 shows how to use this equation.

WORKED EXAMPLE 21.3

A school power pack has an output voltage of 9 V. It is plugged in to the 230 V mains supply. The power pack contains a transformer. The output current of the power pack is 3 A. Calculate the current supplied to the primary coil of the transformer in the power pack. Assume there are no energy losses in the transformer.

Step 1: Draw a transformer symbol and mark on the information from the question:

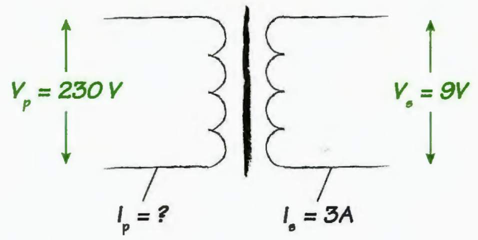


Figure 21.22: Transformer symbol with the known quantities.

Step 2: Write down the transformer power equation.

$I_p \times V_p = I_s \times V_s$

Step 3: Substitute values from the question.

$3\text{ A} \times 9\text{ V} = I_s \times 230\text{ V}$

Step 4: Rearrange and solve for I_s.

$$I_s = \frac{3\text{ A} \times 9\text{ V}}{230\text{ V}}$$
$$= 0.12\text{ A}$$

CONTINUED

Answer

So, the current supplied to the primary coil is 0.12 A.

In stepping down the voltage, the transformer has stepped up the current. If both had been stepped up, we would be getting something for nothing – which in physics, is impossible!

Questions

14 A power station generates 230 000 W.

- **a** Calculate the current in the wires when the power is transmitted:
  - **i** at 230 V
  - **ii** at 23 000 V.
- **b** Explain why it is better to transmit the electricity at a higher voltage.

15 The power supply to a factory is 100 kW. The wires have a resistance of 0.2 Ω.

- **a** Calculate the power loss in the cables when the voltage across the wires is 250 V.
- **b** Calculate the power loss when the voltage is stepped up to 12 500 V.

16 A transformer reduces the mains voltage from 240 V to 12 V for use in a school laboratory power pack. The current supplied by the power pack is 2 A. What current flows into the power pack? What assumption must you make in this calculation?

REFLECTION

Look back through your work on the motor effect and electromagnetic induction. Are there any areas you need to work on? Ask yourself:

- Are you clear about what each effect is?
- Can you recall and understand the equations?
- Can you recall and use the relevant rules for finding directions of fields or currents?

Make revision cards with the key things you need to remember.

PROJECT

Power to the people

Figure 21.23: A town in Sweden.

In this Swedish town coal is mined then burned to generate electricity. The electricity is distributed to homes via the national grid.

Electrical power generation and distribution varies depending on where you live.

Your task is to research how electricity reaches you. Working in a small group you will research how electricity is generated and distributed in your country. You will work together to produce a large poster illustrating and explaining what you find. Your poster will have a lot of information, so plan carefully how the different parts will come together.

For example, you may decide to structure your poster around a diagram of the national grid, with information boxes for each stage.

You must include:

- information about the type of power stations and renewable resources which are in use
- a diagram of a generator and information about how it works and how it is used in power generation
- a diagram similar to Figure 21.16 showing how electricity is distributed
- information about the role of transformers in the system and the difference between step-up and step-down transformers
- reasons why the voltage of the supply has to be changed.

You could add:

- a pie chart showing the different energy resources used in your country
- a detailed explanation of how an a.c. generator works, including why the current generated is a.c. (not d.c.)
- details of the voltages at different parts of the grid, with calculations showing the transformers needed at each stage
- photographs from your local area showing pylons, substations, overhead cables and safety notices.

SUMMARY

When there is relative movement between a conductor and a magnetic field, an e.m.f. is induced across the conductor.

The induced e.m.f. may cause an induced current to flow.

The e.m.f. induced in a coil of wire can be increased by:

- increasing the magnetic field strength
- increasing the number of turns on the coil
- increasing the speed of movement.

The direction of the induced e.m.f. opposes the movement which causes it.

CONTINUED

The direction of the induced e.m.f can be found using Fleming's right-hand rule.
Direct current flows in one direction. Alternating current reverses direction repeatedly.
The output of an a.c. generator depends on the position of the coil in relation to the magnetic field.
A transformer, consisting of a primary coil, a secondary coil and a soft iron core can be used to change alternating voltages.
A step-up transformer increases voltage, a step-down transformer decreases voltage.
The number of turns on transformer coils and the voltages across them can be calculated using the equation. $\frac{V_p}{V_s} = \frac{N_p}{N_s}$
Transformers increase voltage for transmission by the national grid then reduce the voltage to a safer level for consumers.
a.c. in the primary core of a transformer creates a changing magnetic field which, in turn, induces a.c. in the secondary coil.
When a transformer is assumed to be 100% efficient, the primary and secondary currents and voltages are related by the equation $I_p \times V_p = I_s \times V_s$.
The power losses in cables can be calculated using the equation $P = I^2R$.

EXAM-STYLE QUESTIONS

1 Which one of these is not needed for electromagnetic induction? **[1]**

A a conductor
B movement
C a coil of wire
D a magnetic field

2 Which statement is true? **[1]**

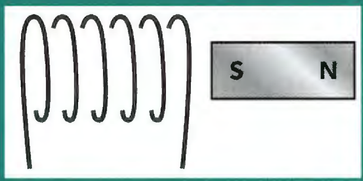
A A step-down transformer decreases current.
B A step-up transformer increases voltage.
C A step-up transformer is used in a school laboratory power supply.
D A step-down transformer has more turns on the secondary coil than on the primary coil.

3 A student investigates the effect of moving a bar magnet into a coil.
She measures the current which flows in the coil when she moves the magnet.
Which change would increase the current? **[1]**

A using a stronger bar magnet
B moving the magnet in the opposite direction
C moving the coil instead of the magnet
D moving the coil from side to side as well as in and out of the magnet

CONTINUED

4 The diagram shows a coil and a magnet. When the magnet is moved away from the coil, current flows.



Which one of the following describes what happens? [1]

A Current flows clockwise in the coil, creating a north pole which attracts the south pole of the bar magnet.

B Current flows anticlockwise, creating a north pole which repels the south pole of the bar magnet.

C Current flows anticlockwise, creating a south pole which repels the south pole of the bar magnet.

D Current flows anticlockwise, creating a north pole which attracts the south pole of the bar magnet.

5 **a** Which statement describes electromagnetic induction? [1]

A the production of an e.m.f across an electrical conductor when there is relative movement between the conductor and a magnetic field

B the production of an e.m.f across an electrical conductor when there is no movement between the conductor and a magnetic field

C the production of an e.m.f across an electrical conductor when there is relative movement between the conductor and an induced current

D the production of an e.m.f across an electrical conductor when there is no movement between the conductor and an induced current

b Describe an experiment to demonstrate electromagnetic induction using a horseshoe magnet, a piece of copper wire and a sensitive ammeter. You may include a diagram in your answer. [3]

c State **two** factors which affect the size of the current induced in this experiment. [2]

[Total: 6]

6 **a** Draw a diagram of a step-up transformer and label the three essential parts of the transformer. [4]

b A student connects the input of a step-up transformer to a battery. He connects the output to a bulb, but it does not light. Explain why. [1]

[Total: 5]

COMMAND WORDS

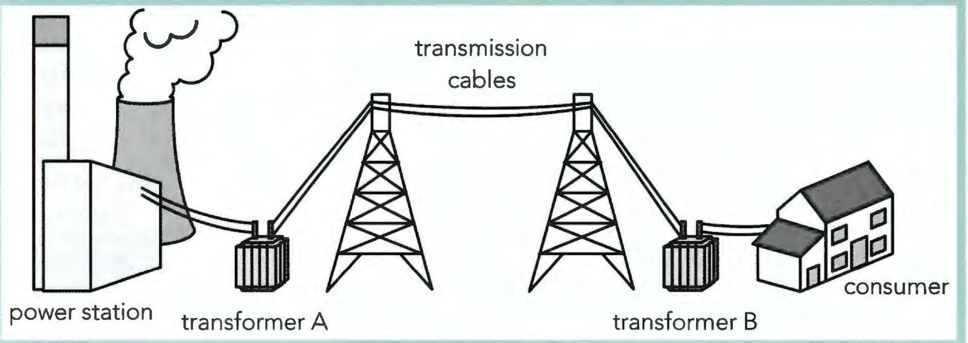
describe: state the points of a topic; give characteristics and main features

state: express in clear terms

explain: set out purposes or reasons; make the relationships between things evident; provide why and/or how and support with relevant evidence

CONTINUED

7 a Transformers are used in the national grid to change the voltage of the supply. For transformers A and B, state whether the transformer is step-up or step-down, and explain why the voltage change is necessary. [4]



b A transformer in the national grid has 800 turns on the primary coil and 16 000 on the secondary. The primary voltage is 25 kV. **Calculate** the secondary voltage. [2]

[Total: 6]

8 A teacher demonstrates electromagnetic induction using a coil, a bar magnet and a sensitive, centre-zero ammeter.

a The teacher slowly moves the magnet towards the coil. State and explain what happens. [2]

b The teacher changes different factors in the experiment. State the effect of each of these actions on the ammeter reading:

i The teacher holds the coil still.

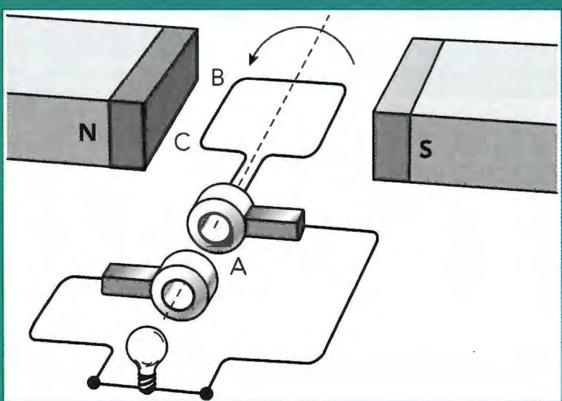
ii The teacher moves the coil quickly towards the coil.

iii The teacher holds the magnet still and slowly moves the coil towards the magnet.

iv The teacher quickly moves the magnet away from the coil. [4]

[Total: 6]

9 The diagram shows an a.c. generator.



COMMAND WORDS

calculate: work out from given facts, figures or information

CONTINUED

a Name the part labelled A. [1]

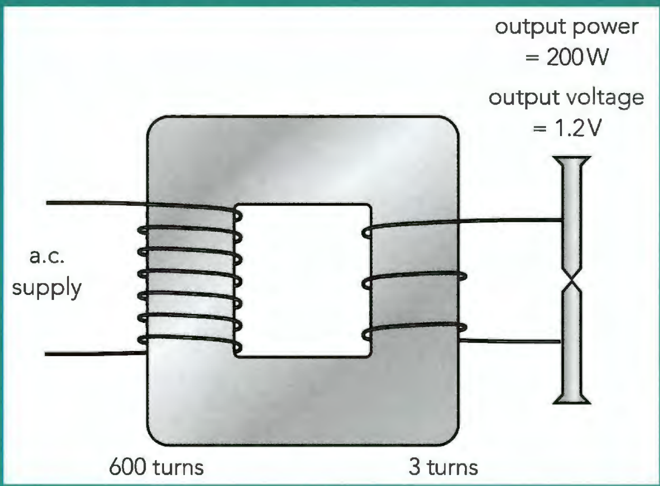
b Name the rule which is used to find the direction of the current in a conductor moving in a magnetic field. [1]

c Use this law to find the direction of the force on the section of wire labelled BC. [1]

d The brightness of the lamp varies as the coil is turned. Explain why the position of the coil affects the brightness of the bulb. [3]

[Total: 6]

10 Spot welding is a method of joining two pieces of metal together by passing a very high current through them. The diagram shows a transformer being used to spot weld two iron nails.



a Explain why the nails become hot when the power is switched on. [1]

b Use the data from the diagram to calculate the current needed from the power supply. Assume the transformer is 100% efficient. **Give** your answer to two decimal places. [4]

[Total: 5]

COMMAND WORDS

give: produce an answer from a given source or recall / memory

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics. This will help you to see any gaps in your knowledge and help you to learn more effectively.

I can	See Topic...	Needs more work	Almost there	Confident to move on
State that an e.m.f. is induced when there is relative movement between a conductor and a magnetic field, and describe an experiment to demonstrate this.	21.1			
List the ways in which the induced e.m.f. can be increased.	21.1			
State that the induced e.m.f. opposes the change which causes it.	21.1			
Use Fleming's right-hand rule to find the direction of the induced current.	21.1			
Describe the structure and operation of an a.c. generator.	21.1			
Sketch a graph of voltage against time for an a.c. generator. State what position the coil must be in for the peaks, troughs and zeros on the graph.	21.1			
List the three essential parts of a transformer.	21.2			
Describe the difference between a step-up and a step-down transformer, by comparing their primary and secondary coils.	21.2			
Recall and use the equation $\frac{V_p}{V_s} = \frac{N_p}{N_s}$.	21.2			
Describe how and why transformers are used in high voltage transmission.	21.2			
Explain how a transformer works.	21.3			
Recall and use the equation $I_p \times V_p = I_s \times V_s$.	21.3			
Recall and use the equation $P = I^2R$.	21.3			